

Maxence Caucheteux

École des Ponts et Chaussées | Master El Karoui

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EDUCATION

École Polytechnique – Sorbonne Université

Msc "Probabilité et Finance" | Dea El Karoui

Paris, France

Sep. 2025 – Present

- **Relevant Coursework:** Stochastic Modeling of Derivatives (E.Abi Jaber), High Frequency Trading (M.Rosenbaum), Rate Models (N.El Karoui), Stochastic Calculus (N.Broutin), Numerical Methods (G.Pagès), Stochastic Control (I.Kharroubi), Machine Learning and Deep Learning (P.Gallinari, B.Wilbertz), Risk Measures (A.Alfonsi)

École Nationale des Ponts et Chaussées

Master of Engineering (Probability and Finance) – Grande École Program

Paris, France

Sep. 2023 – Present

- **Relevant Coursework:** Applied Mathematics (Stochastic Processes, Stochastic Calculus, Advanced Statistics, Partial Differential Equations, Optimisation, Control Theory, Functional Analysis), Computer Science (Programming, Machine Learning, Deep Learning), Finance (Financial Markets)

Lycée Louis-Le-Grand

Preparatory Classes (MP2I & MPI*)

Paris, France

Sep. 2021 – Jul 2023

- Intensive preparation for the nationwide competitive entrance exam into French "Grandes Écoles" Engineering Schools
- **Coursework:** Mathematics, Physics, Computer Science, Philosophy, Literature, English
- Ranked 10/39 in Mathematics and Physics in second year
- Accepted to the Mines-Ponts and Centrale-Supélec competitive entrance exams

WORK EXPERIENCE

Barclays

Quantitative Analyst Intern

London, United Kingdom

Apr. 2026 – Oct. 2026

- Internship within the XVA team

University of Paris

Assistant Lecturer in Mathematics

Paris, France

Sep. 2024 – Jul. 2025

- Taught two hours of mathematics classes per week to 40 second-year management students as part of the Quantitative Techniques for Economics course
- Designed exam problems and graded examinations on four occasions

PROJECTS

State spaces of multifactor approximations of Volterra processes

Jun. 2025 – Aug. 2025

- Summer internship supervised by Aurélien Alfonsi. Reference available upon request.
- Studied the state spaces of multifactor Markovian approximations of nonnegative Volterra processes, and extended existing results from the literature (Abi Jaber)
- Explored applications in simulation schemes and PDE methods for nonnegative Volterra processes

Currency option pricing and hedging under Garman-Kohlhagen model

Mar. 2025 – Apr. 2025

- Project on FX options using the Garman-Kohlhagen model: pricing and delta computation for European options, delta hedging simulations, and numerical solution of the pricing PDE for American puts using the Brennan-Schwarz algorithm

Calibration methods for financial risk models

Feb. 2025 – Jun. 2025

- Project proposed by Milliman, exploring the effectiveness of maximum likelihood estimation versus generalized method of moments in calibrating financial risk models

Mathematical modeling of epidemiology and mobility

Jan. 2024 – Jun. 2024

- Academic project with researcher in Mathematics Jean-François Delmas focusing on the influence of population movements in the spread of epidemics, first through stochastic models and then through deterministic models

Classification of integer simplexes

Sep. 2022 – Jun. 2023

- Research project in Discrete Geometry: Construction of a classification of integer simplexes based on their number of integer points, with an emphasis on algorithms to compute this classification

EXTRACURRICULAR ACTIVITIES

Ponts AI

École nationale des ponts et chaussées

Paris, France

May. 2024 – May. 2025

- Secretary General of a Student Artificial Intelligence Association
- Part of a team of 6 students organizing events aimed at fostering AI development within the student community
- Involved in events organized by the association with Meta, the Australian Government, Capgemini, Instadeep, EffiSciences (French competitiveness cluster focusing on innovative technologies), Science Trash (scientific YouTuber with 750k+ subscribers)

Math Club

École nationale des ponts et chaussées

Paris, France

Sep. 2024 – Jun. 2025

- Active member of the School Association
- Proposed solutions to mathematical problems presented by the club, related to quotient groups and normal subgroups

SKILLS AND INTERESTS

Programming: Python, C/C++, Julia, OCaml

Languages: French (Native), English (Advanced), Spanish (Conversational)

Hobbies: Running (5 times a week, 5k personal best: 16:56, 10k personal best: 37:16), Writing